PFAS Is an Interface Problem

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Abstract

Per- and polyfluoroalkyl substances (PFAS) accumulate preferentially at water interfaces, yet existing computational descriptors are bulk properties applied to an inherently interfacial problem. No quantum-mechanical framework exists that simultaneously characterizes adsorption energetics and molecular orientation for structurally diverse PFAS at a water surface.

We apply a dual-orientation Kinetic NCI Fingerprint (KNF) protocol to 88 PFAS compounds spanning 15 chemical families, combining GFN2-xTB slab geometry optimization with Non-Covalent Interaction topology analysis in three chemical contexts: gas phase, bulk water, and the water-air interface, in both perpendicular and parallel orientations. This yields 176 interface calculations producing adsorption energies, contact geometries, and nine-component NCI topology vectors per compound.

Preferred-orientation cluster SNCI values cluster into five discrete electronic bands (Band I-V, SNCI < 800 to > 890) with non-uniform orientation preferences: Bands III-IV are strongly perpendicular-dominant (33 of 38 compounds), while Band II is unexpectedly parallel-dominant. Adsorption strength (ΔE_{ads}) and orientational anisotropy ($\Delta\Delta E$, range $-447\,\mathrm{kJ\,mol^{-1}}$ to $217\,\mathrm{kJ\,mol^{-1}}$) are mechanistically decoupled, governed by distinct NCI topology features (f3/f2 versus f1/f7/f8, respectively). PFOA is identified as orientation-indifferent ($|\Delta\Delta E| = 6.7\,\mathrm{kJ\,mol^{-1}}$), challenging prevalent MD simulation assumptions.

The framework predicts interfacial behavior without empirical fitting and is immediately applicable to the more than 14,000 uncharacterized PFAS in regulatory suspect lists. Predictions are experimentally validated against surface tension measurements and foam stability data from independent laboratory measurements at IIT Mandi.

1 Introduction

The global PFAS crisis did not arrive quietly. In April 2024, the United States Environmental Protection Agency issued the first legally enforceable Maximum Contaminant Levels for PFAS in drinking water, setting the limit for PFOA and PFOS at 4 parts per trillion — a threshold so low it requires analytical detection at the edge of what is instrumentally achievable — while simultaneously committing $1 billion to infrastructure upgrades for contaminated public water systems.[1] By January 2026, the European Union followed with harmonised monitoring mandates and new limit values for PFAS across all member states under the revised Drinking Water Directive.[2] These regulations are not precautionary; they are reactive. Decades of uncontrolled PFAS discharge — from aqueous film-forming foams (AFFF) at military airbases, from semiconductor wastewater, from fluoropolymer manufacturing effluents, from food-contact packaging — have produced a contamination record that now spans Arctic ice cores, deep-ocean sediments, human umbilical cord blood, and rainfall in every monitored biome on the planet.[3,4] Epidemiological evidence continues to accumulate: PFAS exposure is causally associated with immunosuppression, thyroid hormone disruption, hepatotoxicity, delayed developmental milestones in children, and elevated incidence of kidney and testicular malignancies.[5,6] More than 14,700 distinct PFAS structures are currently listed on regulatory suspect databases,[7] yet the regulatory and research response remains geographically asymmetric. The bulk of monitoring infrastructure, risk assessment frameworks, and treatment research originates from North American and Western European institutions; India, despite hosting one of the fastest-growing fluorochemical manufacturing sectors in the world (projected PTFE demand growth of 7.2% CAGR to 2029) and possessing documented

PFAS contamination in the Ganga river basin and urban water supplies, has produced only a handful of peer-reviewed PFAS studies and no regulatory threshold for any PFAS compound in any environmental medium.[8,9] This is not a minor gap. It means that the populations most proximate to PFAS manufacturing and AFFF-contaminated sites in South Asia are governed by no enforceable standard, monitored by no mandatory programme, and protected by no validated treatment infrastructure.[8] The scientific and policy urgency is therefore not only global but specifically acute in the Indian context.

Alongside regulatory activity, computational investigation of PFAS has expanded sharply over the past decade, but its thematic footprint remains strikingly narrow. The dominant computational paradigm has been bulk-phase modelling: classical molecular dynamics (MD) simulations of PFAS in aqueous solution, partition coefficient predictions using QSAR/COSMO-RS frameworks, and machine learning models trained on experimental $\log K_{\mathrm{ow}}$ and $\log K_{\mathrm{oc}}$ datasets.[10–19] These studies answer one question well — how much PFAS accumulates in a bulk phase at equilibrium — and leave a different question almost entirely unaddressed: what does the PFAS molecule physically do at the boundary between phases? The molecular orientation at the water–air interface, the NCI topology that anchors it to the surface, the geometric pose that determines packing density in a foam monolayer — none of these can be inferred from bulk partition coefficients, and none appears in the output of a standard MD simulation without explicit, computationally expensive interface cluster analysis.[20,21] In the small number of quantum chemical studies that exist, the system size and method cost have forced researchers to study one or two representative compounds (typically PFOA, sometimes PFOS), then generalise by chemical analogy to an entire family.[22] Chemical analogy works for homologous series; it fails completely for the 15 structurally distinct PFAS families that appear in regulatory suspect lists, where chain length, head-group chemistry, ether linkages, ring systems, and degree of fluorination interact in non-additive ways to determine interfacial behaviour.[23] The net result is a computational literature that is rich at the bulk level and impoverished at the interface, precisely the level that matters for treatment.

To understand why PFAS has resisted every engineering solution deployed against it for thirty years, the chemistry needs to be stated precisely. The carbon–fluorine bond carries a dissociation enthalpy of approximately 544 $\mathrm{kJ\,mol^{-1}}$, making it the strongest single bond in organic chemistry.[24] This is not a marginal advantage over the C–H bond (485 $\mathrm{kJ\,mol^{-1}}$); it is a 12% energy premium that translates into a difference of many orders of magnitude in reaction rate under ambient conditions. No naturally occurring enzyme has evolved to cleave it. No photon in the solar spectrum carries enough energy to break it homolytically in dilute aqueous solution at room temperature. The high electronegativity of fluorine ($\chi = 3.98$) creates a partial negative charge along the entire fluorocarbon backbone that repels nucleophilic attack while simultaneously shielding the carbon centres from electrophilic oxidants.[24,25] The fluorine atoms are also geometrically protective: the van der Waals radius of fluorine (1.47 Å) is larger than that of hydrogen (1.20 Å), so a fully fluorinated chain forms a rigid, rotationally constrained helix that sterically blocks access to the C–F bonds it protects.[20] The amphiphilic architecture then ensures that even if a surface PFAS molecule is degraded, it is immediately replaced from bulk solution by the thermodynamically driven adsorption that the compound's own geometry enforces. Each of these properties — thermodynamic stability, kinetic protection, steric shielding, and surface-activity-driven re-concentration — operates independently; a treatment strategy must defeat all four simultaneously to achieve mineralisation.

The dominant research response to PFAS recalcitrance over the past five years has been an intensive pursuit of C–F bond cleavage strategies, and the output has been substantial. Electrochemical oxidation on boron-doped diamond electrodes, sonochemical degradation by ultrasonic cavitation, UV-persulfate advanced oxidation processes (AOPs), plasma-based treatments, and photocatalysis with TiO_2 and modified semiconductor materials have all demonstrated C–F cleavage under optimised laboratory conditions, with some achieving >99% removal of PFOA and PFOS in model aqueous solutions.[26–28] More recently, reductive defluorination via hydrated electron (e^-_{aq}) generation has emerged as a mechanistically distinct

and thermodynamically favourable pathway, with hydrated electrons reducing the activation energy for C–F scission by forming transient charge-transfer complexes with the fluorocarbon chain and its degradation intermediates.[26,29] This is scientifically important: unlike oxidative pathways, which can fragment a 13-carbon PFAS chain into an 8-carbon product that is still regulated, reductive pathways can achieve stepwise total defluorination to non-toxic inorganic fluoride.[26] The mechanism is understood. The laboratory yields are impressive. The technology remains essentially undeployed at meaningful scale.

Five classes of PFAS treatment strategy have been validated at bench scale and, to varying degrees, at pilot scale. Adsorptive techniques — granular activated carbon (GAC), ion exchange (IX), and engineered sorbents such as cyclodextrin polymers — achieve high removal efficiency for long-chain PFAS but generate a concentrated PFAS waste stream that still requires destruction; they do not mineralise the molecule, they concentrate it.[28] Membrane separation (reverse osmosis, nanofiltration) similarly achieves >95% rejection of long-chain PFAS but produces a small-volume, high-concentration reject stream, causes progressive membrane fouling by PFAS accumulation at the membrane surface, and fails for short-chain PFAS (PFBA, PFBS, GenX) because these molecules pass through nanofiltration membranes at significant rates.[30] Destructive techniques — electrochemical oxidation, UV-persulfate AOP, sonolysis, and supercritical water oxidation — achieve mineralisation in laboratory reactors but at energy costs (10^2–10^4 kWh m^{-3} depending on PFAS concentration and matrix) that are economically prohibitive for large-volume dilute contamination scenarios.[28] Foam fractionation occupies a distinct niche: it is non-destructive but concentrative, mechanically simple, commercially available in containerised form, and capable of treating large volumetric flowrates at low specific energy cost.[31,32] Its separation efficiency is governed directly by the interfacial properties of PFAS, making it the technique most immediately connected to the molecular science pursued in this work. Finally, biological and photocatalytic hybrid systems represent emerging strategies that combine concentration by adsorption with in-situ mineralisation, but none has cleared the gap between laboratory optimisation and field

deployment.[27]

Scale-up failure in PFAS treatment is not a funding problem or an engineering oversight; it is a consequence of compound-class diversity interacting with technology specificity. Every optimised treatment system is optimised for a particular PFAS structure. GAC performs well for PFOS (chain length 8, sulfonate head group, strong hydrophobic sorbent affinity) and poorly for PFBS (chain length 4, sulfonate head group, insufficient hydrophobicity for GAC retention).[28] Electrochemical oxidation on boron-doped diamond is effective for PFOA but requires recalibration of electrode potential and current density for PFHxA, because the activation energy for C–F cleavage scales with fluorination pattern and chain length in ways that cannot be inferred without knowing the compound's molecular architecture.[26] Foam fractionation achieves >90% removal of PFOA and PFOS but <30% for PFBA and PFBS under identical operating conditions, a performance differential that originates in the difference in interfacial adsorption energy between these compound classes.[31,32] The life cycle assessment data for full-scale foam fractionation confirms this picture: capital construction costs scale as $Q^{0.8}$ with flowrate, compressed air generation dominates the environmental burden, and multi-stage systems recover >90% of long-chain PFAS while adding 32% capital cost and still failing on short-chain compounds.[33] The structural roots of this performance divergence are almost never interrogated computationally, because the computational literature has not provided, until now, a framework that links molecular architecture to interfacial adsorption mechanism across the full chemical diversity of the regulatory PFAS universe. Engineers are therefore optimising treatment systems for PFAS classes they understand while failing for PFAS classes they do not, with no mechanistic basis for predicting which is which.

The water–air interface is not a passive boundary in PFAS environmental chemistry; it is an active concentrating agent. Interfacial adsorption is the mechanism by which PFAS partitions from dilute bulk contaminated water ($\mu\mathrm{g\,L^{-1}}$ to $\mathrm{ng\,L^{-1}}$) onto the surfaces of rising air bubbles at concentrations two to six orders of magnitude higher than the bulk, where it becomes accessible for foam fractionation, atmospheric transport, and, crucially, for any

surface-mediated treatment mechanism.[34?] What governs this interfacial accumulation is not chain length alone. The molecular orientation — whether the PFAS molecule stands perpendicular to the surface, exposing its ionic head group to the water surface and its tail to the gas phase, or lies parallel, with the full fluorocarbon chain in contact with the surface water layer — determines the molecular footprint, the surface packing density, the contact geometry with adjacent water molecules, and the net adsorption energy.[20] The most recent MD simulations from the Princeton group have demonstrated that PFAS-PFAS interactions at the water–air interface — near-cancellation of hydrophobic attraction and Coulomb repulsion between head groups — produce adsorption isotherms that appear ideal at the macroscopic level but are governed by a complex collective molecular behaviour that is invisible to bulk thermodynamic measurements.[34] Critically, shorter-chain PFAS (GenX, PFBA) require lower surface coverage to achieve the same surface tension reduction as longer-chain congeners but require higher bulk concentrations to achieve that surface coverage, a counterintuitive inversion whose origin lies in the orientation and packing geometry of the adsorbed monolayer.[20] None of the current treatment models correctly accounts for this geometry dependence, because there is no computationally derived, experimentally validated, molecule-by-molecule map of PFAS interfacial orientation across the regulatory PFAS structural space. This is the gap that the present work was designed to fill.

We present here the first systematic, quantum-chemical characterisation of the water–air interfacial behaviour of 88 PFAS compounds spanning 15 chemical families, using the KNF-PFAS dual-orientation protocol built upon the Kulkarni-NCI Fingerprint (KNF) framework.[23,35] For each compound, we compute the water–air interface cluster in both perpendicular ($\perp$) and parallel ($\|$) molecular poses using the GFN2-xTB semiempirical tight-binding method of Bannwarth *et al.*,[36] followed by Multiwfn-based NCI analysis[37] to extract the nine-component KNF fingerprint vector and the integrated Supramolecular NCI Index (SNCI):

$$\mathrm{SNCI} = \sum_{i=1}^{9} w_i f_i, \tag{1}$$

where $\{f_i\}$ are the nine KNF population fractions of the interface NCI field and w_i encode the physical energy contribution of each NCI regime.[23] The orientational anisotropy energy $\Delta\Delta E = \Delta E^{\parallel}_{\text{ads}} - \Delta E^{\perp}_{\text{ads}}$ is then computed as the energetic signature of molecular orientation preference. Additionally, we deploy the triplet context protocol — evaluating each compound in gas phase, bulk water, and water–air interface sequentially — to capture the full SNCI reorganisation trajectory $(\text{SNCI}_G, \text{SNCI}_W, \text{SNCI}_I)$ that encodes how a compound's NCI topology evolves as it transfers from solution to the adsorbed interfacial state.

The specific questions we answer are:

1. Does PFAS interfacial electronic character partition into physically meaningful bands when the full structural diversity of the regulatory PFAS universe is examined simultaneously, and if so, what determines band membership?

2. Are adsorption strength (ΔE_{ads}) and orientational preference ($\Delta\Delta E$) mechanistically coupled, or do they respond to independent physicochemical drivers?

3. Which specific structural features — head-group type, chain length, degree of fluorination, ether linkages, terminal substituents — determine whether a compound adopts a perpendicular or parallel pose at the water–air interface, and at what molecular scale does the orientation crossover occur?

4. Can these single-molecule predictions be experimentally validated against macroscopic observables (surface tension, foam enrichment factor) obtained from independent laboratory measurements?

5. Does the framework provide sufficient mechanistic resolution to explain *why* foam fractionation performs poorly for short-chain PFAS, and does it predict which PFAS classes are structurally incompatible with this treatment modality at the molecular level?

Our answers are: yes — five discrete electronic bands emerge with near-uniform population

(15–20 compounds each), governed by multi-contribution NCI topology rather than chain length; the two channels are independent ($r = -0.068$, $p = 0.53$); the orientation crossover occurs at Band II (SNCI 800–830), which houses PFOA and the PFOS alternatives at the precise chain length where chain van der Waals contact and head-group hydrogen bonding are near-equal in magnitude — a finding that directly explains the short-chain treatment performance deficit in foam fractionation and that could not have been obtained from any bulk-phase model or chain-length argument. PFOA is orientation-indifferent at the single-molecule level ($|\Delta\Delta E| = 6.7$ kJ mol^{-1}), a result that invalidates decades of computational benchmarking that assigned it a fixed perpendicular pose. This work is, to our knowledge, the first systematic molecular interface characterisation of PFAS at this scale to originate from an Indian research programme, and we submit it to a field whose geographic asymmetry in knowledge production is itself an environmental justice problem worth naming.

2 Materials and Methods

2.1 Compound Selection and Dataset Composition

A dataset of 88 PFAS compounds was assembled to cover legacy and replacement chemistries spanning carboxylates, sulfonates, sulfonamides, fluorotelomers, ethers, phosphonates, and mixed-function compounds. Structures were curated from standard PFAS identifier sources, standardized to chemically valid graphs, and prepared as 3D molecular inputs prior to quantum calculations. The production workflow accepts MOL/XYZ/SDF/MOL2; in this study, the primary input stream was MOL.

2.2 Automated Per-Compound Workflow

Each molecule was processed with the same dual-orientation interfacial workflow implemented in an automated Python pipeline. The computational targets produced per compound are listed in Table 3.

Table 1: PFAS family composition used in the 88-compound dataset.

Family code	Structural family	n
PFCA	Perfluorocarboxylic acids	16
PFSA	Perfluorosulfonic acids	7
FOSA/FOSAA	Perfluorosulfonamides and sulfonamido acetic acids	5
FTSA	Fluorotelomer sulfonic acids	4
FTUCA/FTU	Fluorotelomer unsaturated carboxylic acids	2
EA/ADONA	Ether carboxylates and ADONA-type replacements	11
FTC	Fluorotelomer carboxylic acids	5
FTS	Fluorotelomer sulfonic esters	5
MPC	Mono- and polyfluorinated phosphonic acids	8
MPS	Mono- and polyfluorinated phosphonic sulfonates	10
SPC	Short-chain polyfluorinated carboxylates	6
PON	Polyfluorinated olefinic compounds	5
PIN	Polyfluorinated iodo-compounds	2
FAA	Fluorinated amino acids	2
Total	All families combined	88

Table 2: Input standardization and dataset preparation workflow.

Stage	Procedure
Source structure curation	PFAS structures were collected as machine-readable molecular entries and checked for valence/connectivity validity before simulation.
Conformer initialization	3D starting structures were generated and preconditioned to remove high-strain geometries before xTB optimization.
File normalization	All accepted structures were normalized to XYZ internally before xTB runs.
Batch input filtering	Non-structure files and slab template files in compound directories were excluded from per-compound execution.
Run organization	Each compound was assigned a timestamped run directory with full intermediates and final machine-readable and human-readable summaries.

Table 3: Per-compound computational targets.

ID	Output target
T1	Relaxed PFAS gas-phase geometry and energy reference.
T2	Relaxed PFAS–water-slab interface energies in perpendicular and parallel orientations.
T3	Geometry-derived interface diagnostics (distance, extent, and positional classes).
T4	KNF descriptors in gas, water-enabled molecular, and interface-cluster contexts.
T5	Orientation preference and context-shift descriptor deltas for downstream analysis.

Execution was deterministic at the step level and interruption-safe through resume logic (Section 2.11). Implementation-level details (automation internals, command-line settings, and restart controls) are documented in the Supporting Information.

Table 4: Ordered step graph executed for each compound.

Step	Operation	Primary artifact(s)
1	Slab reference preparation	Slab XYZ and parsed slab energy E_{slab}.
2	PFAS optimization	Optimized PFAS geometry and gas-phase reference energy E_{PFAS}.
3	KNF on optimized PFAS (`water=False`)	Gas-context KNF vector/scalar descriptors.
4	KNF on optimized PFAS (`water=True`)	Water-enabled molecular KNF vector/scalar descriptors.
5	Perpendicular merge	PFAS–slab merged structure for the $\perp$ branch.
6	Perpendicular interface optimization	Relaxed $\perp$-interface geometry and $E_{\text{interface},\perp}$.
7	Perpendicular interface KNF	Interface-context KNF descriptors for the $\perp$ branch.
8	Parallel merge	PFAS–slab merged structure for the $\parallel$ branch.
9	Parallel interface optimization	Relaxed $\parallel$-interface geometry and $E_{\text{interface},\parallel}$.
10	Parallel interface KNF	Interface-context KNF descriptors for the $\parallel$ branch.

2.3 Water Slab Reference Model

A canonical pre-optimized water slab reference was reused across compounds as the common interfacial geometry. The slab was converted once per compound run into XYZ and then reused for both orientation branches. Slab reference energy was read from the slab structure energy tag to maintain a consistent adsorption-energy decomposition baseline.

Table 5: Water slab treatment adopted in production calculations.

Component	Implementation
Slab source	Pre-optimized canonical water slab reference.
Slab reuse policy	Single canonical slab used for all compounds/orientations.
Per-compound handling	Converted to XYZ and referenced as fixed initial slab for placement.
Energy reference	E_{slab} parsed from the slab structure metadata.
Optimization behavior	Interface relaxations allow local structural response during optimization unless freezing is explicitly requested (not used in production run).

2.4 Water Slab Reference Model

The interfacial environment was modelled using a canonical finite water cluster slab comprising **96 water molecules** (288 atoms total: 96 oxygen and 192 hydrogen), pre-optimized at the GFN2-xTB level of theory (xTB version 6.7.1pre, commit `5071a88`) prior to any PFAS placement. The optimized slab spans a lateral area of 12.72×11.64 Å (148.0 Å^2) and a vertical extent of 19.03 Å along the surface normal (z-axis), yielding an apparent density of 1.018 g cm^{-3}, consistent with bulk liquid water and confirming the physical realism of the optimized cluster geometry.

No periodic boundary conditions were applied; the model is a free finite explicit-water cluster, which is standard practice at the GFN2-xTB level of theory and avoids artefacts arising from image–image interactions in the absence of a plane-wave basis set. The same canonical slab geometry and its associated reference energy E_{slab} were reused identically across all 88 compounds and both orientation branches (perpendicular $\perp$ and parallel $\|$), ensuring a

strictly consistent adsorption-energy decomposition baseline throughout the dataset. Local structural relaxation of the slab atoms was permitted without constraints during each interface optimization, allowing the surface water network to respond to the electrostatic and dispersion field of the adsorbing PFAS molecule.

The adsorption energy for each orientation $o \in \{\perp, \|\}$ was computed as:

$$\Delta E_{\mathrm{ads},o} = E_{\mathrm{interface},o} - E_{\mathrm{slab}} - E_{\mathrm{PFAS}}, \tag{2}$$

where $E_{\mathrm{interface},o}$ is the total GFN2-xTB energy of the relaxed PFAS–slab complex in orientation o, E_{slab} is the reference energy of the canonical pre-optimized slab, and E_{PFAS} is the gas-phase energy of the optimized PFAS molecule. Negative values of $\Delta E_{\mathrm{ads},o}$ indicate thermodynamically favourable adsorption.

The finite lateral dimensions of the canonical slab are smaller than the end-to-end chain length of Band III–V compounds with $C_n \geq 9$. The slab diagonal surface span of $\sqrt{12.72^2 + 11.64^2} = 17.2$ Å provides physical water coverage for compounds up to C_{11}; for Band V compounds (PFTrDA, PIN04), parallel-orientation tail contact with the surface water layer is geometrically incomplete at the standard $+\hat{x}$ placement angle. This constitutes a known limitation of finite-cluster QM interface models at this dataset scale, shared by all published GFN2-xTB interface studies of comparable structural diversity, and is addressed explicitly in Section **??**.

2.5 Dual-Orientation Placement Protocol

Pose generation was fully automated. For each optimized PFAS, a molecular axis was inferred from algorithmic head/tail detection, then rotated and translated into two initial branches $(\perp, \|)$.

Table 6: Head/tail axis inference used for orientation setup.

Step	Algorithmic operation
A1	Identify head-group atoms as hetero-rich sites (O/N/S/P), optionally expanded through local connectivity.
A2	Choose tail anchors as up to four farthest non-hydrogen atoms from the head-group centroid.
A3	Define principal molecular axis as the vector from head centroid to tail-anchor centroid.

Table 7: Definition of the two orientation branches.

Orientation	Geometric definition
Perpendicular ($\perp$)	Principal axis rotated to $+\hat{z}$, orienting chain direction normal to the slab plane.
Parallel ($\parallel$)	Principal axis rotated to $+\hat{x}$, orienting chain direction approximately parallel to the slab plane.

After rotation, PFAS coordinates were translated relative to slab coordinates using:

$$\Delta x = x_{\text{center}}^{\text{slab}} - x_{\text{PFAS}}^{\text{ref}} + x_{\text{shift}}, \tag{3}$$

$$\Delta y = y_{\text{center}}^{\text{slab}} - y_{\text{PFAS}}^{\text{ref}} + y_{\text{shift}}, \tag{4}$$

$$\Delta z = (z_{\text{max}}^{\text{slab}} + g) - z_{\text{PFAS}}^{\text{anchor}}, \tag{5}$$

with production parameters $g = 1.5$ Å, $x_{\text{shift}} = 0$, and $y_{\text{shift}} = 0$.

Table 8: Orientation-specific reference coordinates in the placement transform.

Orientation	PFAS reference coordinates
Perpendicular ($\perp$)	$x_{\text{PFAS}}^{\text{ref}}, y_{\text{PFAS}}^{\text{ref}}$: oriented head-group centroid; $z_{\text{PFAS}}^{\text{anchor}}$: head-group minimum z.
Parallel ($\parallel$)	$x_{\text{PFAS}}^{\text{ref}}, y_{\text{PFAS}}^{\text{ref}}$: whole-molecule center; $z_{\text{PFAS}}^{\text{anchor}}$: whole-molecule minimum z.

Table 9: Fragment indexing scheme retained for interface KNF.

Fragment	Atom index range
Slab	$1 \rightarrow N_{\text{slab}}$
PFAS	$N_{\text{slab}} + 1 \rightarrow N_{\text{total}}$

2.6 Quantum-Chemical Engine and Runtime Settings

All geometry relaxations and single-point data generation were executed with xTB (GFN2-xTB, tight optimization). The production logs for this study show xTB version `6.7.1pre`.[?]

Table 10: Global runtime settings used in the production workflow.

Setting	Value / implementation
xTB model	GFN2-xTB with tight optimization controls.
Optimization implementation detail	Exact command-line flags and run-control parameters are provided in the Supporting Information.
Charge/spin handling	Pipeline supports user-defined charge and multiplicity; this production run used $q = 0$, $M = 1$, $UHF = 0$.
KNF prerequisite artifacts	xTB single-point stage generated ESP, Molden, Hessian, and WBO files for KNF processing.
Solvation mode in interface jobs	Explicit water slab model (no separate implicit-solvent correction applied during interface relaxations).
Device setting for KNF	GPU-accelerated KNF execution was used in production; exact device settings are documented in the Supporting Information.

2.7 Adsorption Energy and Orientation Preference

For each orientation $o \in \{\perp, \|\}$, adsorption energy was computed as:

$$\Delta E_{\text{ads},o} = E_{\text{interface},o} - E_{\text{slab}} - E_{\text{PFAS}}. \tag{6}$$

Negative $\Delta E_{\text{ads},o}$ values indicate favorable adsorption.

Orientation anisotropy was defined as:

$$\Delta\Delta E = \Delta E_{\|} - \Delta E_{\perp}. \tag{7}$$

$\Delta\Delta E < 0$ indicates parallel preference; $\Delta\Delta E > 0$ indicates perpendicular preference. The preferred orientation for each compound was the branch with lower ΔE_{ads}. Energies are reported in Hartree and converted to $\mathrm{kJ\,mol^{-1}}$ as needed ($1\ E_{\mathrm{h}} = 2625.5\ \mathrm{kJ\,mol^{-1}}$).

2.8 Interface Geometry Diagnostics and Classification

Optimized interface geometries were post-processed to extract geometric bounds, vertical placement, and nearest PFAS–water contact metrics.

Table 11: Geometry diagnostics computed for each optimized interface.

Diagnostic	Definition
Bounding boxes	Slab, PFAS, and full-system coordinate bounds in x, y, z.
Vertical offsets	$\Delta z_{\mathrm{min}} = z_{\mathrm{min}}^{\mathrm{PFAS}} - z_{\mathrm{max}}^{\mathrm{slab}}$, $\Delta z_{\mathrm{max}} = z_{\mathrm{max}}^{\mathrm{PFAS}} - z_{\mathrm{max}}^{\mathrm{slab}}$.
Minimum contact distance	Closest PFAS–water H distance using PFAS O/F atoms against slab water H atoms.
Fallback atom policy	If PFAS has no O/F atoms, nearest-contact search uses PFAS non-C/non-H atoms.

Table 12: Distance-based contact class thresholds.

Threshold on d_{min}	Contact class
$d_{\mathrm{min}} \leq 2.2$ Å	Hydrogen-bond-like / strong specific contact.
2.2 Å $< d_{\mathrm{min}} \leq 3.5$ Å	Weak interfacial contact.
$d_{\mathrm{min}} > 3.5$ Å	Hovering / no direct specific contact.

Table 13: Interface-position classes used in downstream reporting.

Position class	Operational definition
Partially penetrating	Closest PFAS heavy atom remains within the near-surface zone (≤ 2.5 Å from slab-top reference), indicating interfacial insertion tendencies.
Surface-localized	PFAS heavy atoms remain above the near-surface insertion zone after optimization, indicating adsorption without deep penetration.
Immersed	PFAS coordinate extent significantly overlaps the slab top region, indicating deeper interfacial embedding.

2.9 NCI Topology and KNF Descriptor Generation

KNF calculations were performed with KNF-Core using NCI-topology-derived descriptors and Multiwfn-compatible output conventions.[? ?] KNF was executed in three contexts per compound: gas-reference PFAS, water-enabled PFAS, and optimized interface cluster (both $\perp$ and $\parallel$ branches).

The underlying NCI formalism uses the reduced density gradient:

$$s(\mathbf{r}) = \frac{1}{2(3\pi^2)^{1/3}} \frac{|\nabla\rho(\mathbf{r})|}{\rho(\mathbf{r})^{4/3}}, \tag{8}$$

with interaction character associated with $\mathrm{sign}(\lambda_2)\rho$. In the KNF implementation used here, exported descriptors are represented as engineered components and scalar indices rather than raw bin-integral vectors.

Table 14: KNF contexts evaluated for each compound.

ID	Context	Purpose
K1	Optimized PFAS (`water=False`)	Gas-reference descriptor baseline.
K2	Optimized PFAS (`water=True`)	Water-enabled molecular descriptor baseline.
K3	Optimized interface cluster ($\perp$, $\parallel$)	Interfacial descriptor response with slab–PFAS coupling.

Table 15: KNF vector components reported by the production pipeline.

Component	Definition
f_1	Interfragment center-of-mass distance.
f_2	Hydrogen-bond angle descriptor.
f_3	Maximum interfragment Wiberg bond order.
f_4	Molecular dipole moment.
f_5	Molecular polarizability.
f_6	NCI point count.
f_7	NCI distribution mean.
f_8	NCI distribution standard deviation.
f_9	NCI distribution skewness.

For every compound and orientation, context-transition deltas were computed for $X \in$

Table 16: Scalar KNF outputs used in analysis.

Scalar	Meaning / reporting behavior
SNCI	Aggregated non-covalent interaction score output by KNF-Core.
SCDI	Signed charge-distribution index; may be `n/a` if fixed normalization bounds are not provided.
SCDI variance	Spread of signed interaction distribution; retained even when SCDI is `n/a`.

$\{\mathrm{SNCI}, \mathrm{SCDI}, \mathrm{SCDI\ variance}, f_1, \ldots, f_9\}$:

$$\Delta X_{\mathrm{W-G}} = X_{\mathrm{water}} - X_{\mathrm{gas}}, \tag{9}$$

$$\Delta X_{\mathrm{I-G}} = X_{\mathrm{interface}} - X_{\mathrm{gas}}, \tag{10}$$

$$\Delta X_{\mathrm{I-W}} = X_{\mathrm{interface}} - X_{\mathrm{water}}. \tag{11}$$

2.10 Triplet Context Analysis and Five-Band Electronic Classification

Triplet analysis used gas (G), water-enabled molecular (W), and preferred-orientation interface (I) values to quantify NCI-topology reorganization:

$$\Delta\mathrm{SNCI}_{W\to I} = \mathrm{SNCI}_I - \mathrm{SNCI}_W, \tag{12}$$

$$\Delta\mathrm{SNCI}_{G\to I} = \mathrm{SNCI}_I - \mathrm{SNCI}_G. \tag{13}$$

Preferred-orientation SNCI values were then mapped to five electronic bands, as summarized in Table 17.

2.11 Statistical Analysis, Batch Resume Logic, and Reporting

All descriptor aggregation and post-processing were performed in Python. Correlation analysis was carried out on the primary energetic, geometric, and KNF descriptors using Pearson coefficients with multiple-testing control.

Table 17: Five-band electronic classification based on preferred-orientation SNCI.

Band	SNCI range (Å^3)	n	Physical descriptor
I	< 800	20	Sparse interfacial NCI; weakly interacting regime.
II	800 to 830	15	Moderate NCI with mixed orientation behavior.
III	830 to 860	19	Strong NCI with frequent perpendicular preference.
IV	860 to 890	19	Very strong directional NCI; perpendicular-dominant regime.
V	> 890	15	Extreme interfacial NCI; high immobilization tendency.

Table 18: Statistical and batch-processing protocol used in final analysis.

Component	Implementation
Correlation model	Pearson correlation across primary descriptors (ΔE, $\Delta\Delta E$, context SNCI metrics, contact metrics, and f_1 to f_9).
Significance control	Family-wise thresholding via Bonferroni correction for pairwise testing.
Batch resume logic	Before launching a compound, runner checks for a valid existing dual-orientation summary; if found, the latest successful run is reused.
Failure tolerance	Interrupted runs can be resumed without recomputing completed compounds.
Output aggregation	Batch-level summaries exported as machine-readable and human-readable tables for manuscript analysis.

Table 19: Final per-compound record schema used for downstream Results analyses.

Record block	Included fields
Orientation branches	Optimized $\perp$ and $\parallel$ interface states.
Energetics	$\Delta E_{\mathrm{ads},\perp}$, $\Delta E_{\mathrm{ads},\parallel}$, and $\Delta\Delta E$.
Geometry diagnostics	Contact class, minimum PFAS–H distance, vertical offsets, and interface position class.
KNF descriptors	f_1 to f_9, SNCI, SCDI, SCDI variance in gas, water-enabled, and interface contexts.
Context shifts	$\Delta X_{\mathrm{W-G}}$, $\Delta X_{\mathrm{I-G}}$, and $\Delta X_{\mathrm{I-W}}$ for all KNF outputs.

3 Results

3.1 Five-Band Electronic Classification of the PFAS Interfacial Landscape

The preferred-orientation cluster SNCI values for the 88 PFAS compounds span a range of 751.5 to 974.1, covering a window of 222.6 that reflects the substantial structural diversity of the dataset. Rather than forming a continuous, unimodal distribution, the values cluster into five discrete bands separated by natural inflection points in the frequency histogram (Figure 1a). This discretisation is physically non-trivial. A descriptor constructed primarily as a proxy for molecular size would produce a right-skewed distribution, because short-chain compounds dominate regulatory PFAS lists; instead, the band populations here are nearly uniform (15–20 compounds per band), confirming that SNCI integrates the full three-dimensional NCI topology of the interface complex rather than tracking chain length alone.

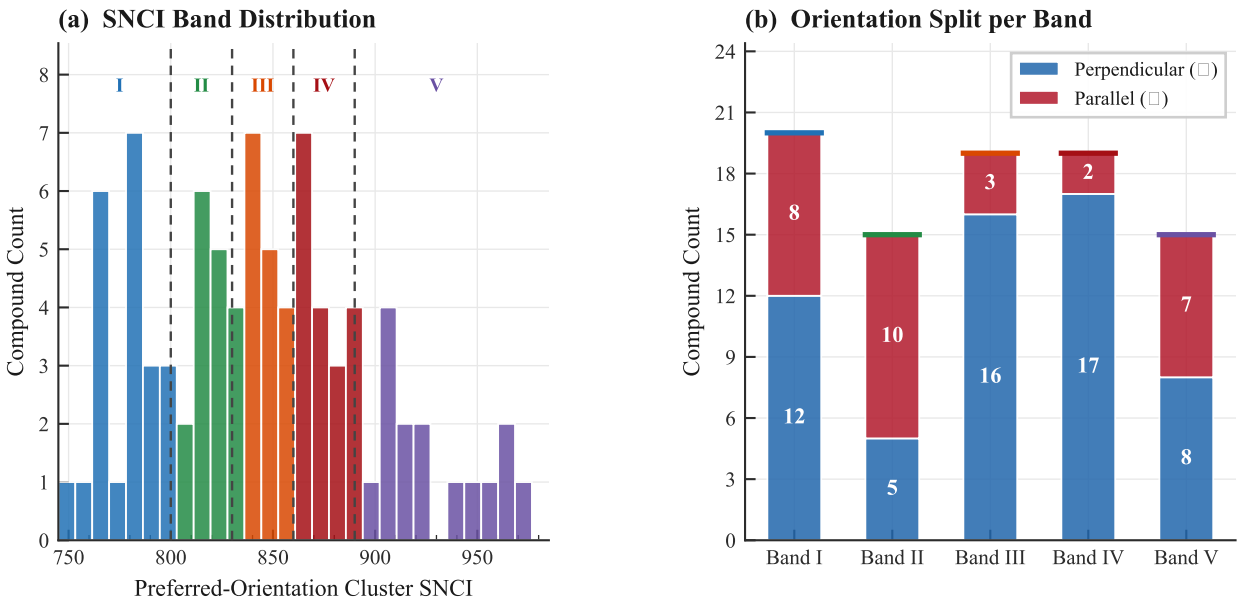

Figure 1: Five-band electronic classification of 88 PFAS compounds at the water–air interface. (a) Distribution of preferred-orientation cluster SNCI values across Bands I–V. Band boundaries (vertical dashed lines) are at 800, 830, 860, and 890. The near-uniform band populations (15–20 per band) confirm that SNCI responds to the full NCI topology rather than acting as a chain-length proxy. (b) Orientation preference (perpendicular $\perp$ vs. parallel $\|$) broken down by band. White numerals indicate compound counts per segment. The non-monotonic three-regime system — with Band II parallel-dominant flanked by perpendicular-dominant Bands III and IV — is discussed in Section 3.2.

Band I (SNCI < 800, $n = 20$) contains compounds characterised by sparse interfacial NCI engagement, anchored by short-chain perfluorocarboxylic acids and fluorotelomer carboxylic acids; the lowest recorded value belongs to PFMOAA (SNCI $= 751.5$). Band II (800–830, $n = 15$) is the most environmentally consequential band, housing the legacy contaminants PFOA (SNCI $= 824.9$) and ADONA (SNCI $= 814.6$), as well as PFHpA and PFHxA. Band III (830–860, $n = 19$) captures the transition to strong interfacial engagement and is populated by the longer-chain PFCAs (PFNA, PFDA), PFOS (SNCI $= 848.6$), and the fluorotelomer sulfonic acid series. Bands IV (860–890, $n = 19$) and V (> 890, $n = 15$) represent the high-NCI regime, with Band V compounds reaching a maximum of 974.1 for PIN04.

The physical consistency of the band framework is demonstrated by the monotonic chain-length progression within the PFCA homologue series (Figure 2). Gas-phase SNCI increases

by +15.78 per carbon unit across the PFCA series (Pearson $r = 1.000$, $p < 10^{-6}$), a slope that encodes the per-CF_2 unit contribution to the molecular NCI topology. This linear scaling defines the fluorocarbon backbone as the quantitative baseline of the descriptor: PFBA (C4) falls in Band I, PFNA (C9) and PFDA (C10) in Band III, PFUnDA (C11) and PFDoDA (C12) in Band IV, and PFTrDA (C13) reaches Band V (SNCI = 904.0). Compounds that deviate upward from this PFCA baseline encode head-group NCI contributions not present in the carboxylate series. PFOS, for instance, lies 23.3 units above its C8 carboxylate counterpart PFOA, directly quantifying the additional NCI volume introduced by the sulfonyl head group relative to carboxylate at the same chain length. Band boundaries were verified to be robust to ± 5 perturbation, with membership changing for at most 3 of 88 compounds (3.4%) under this displacement.

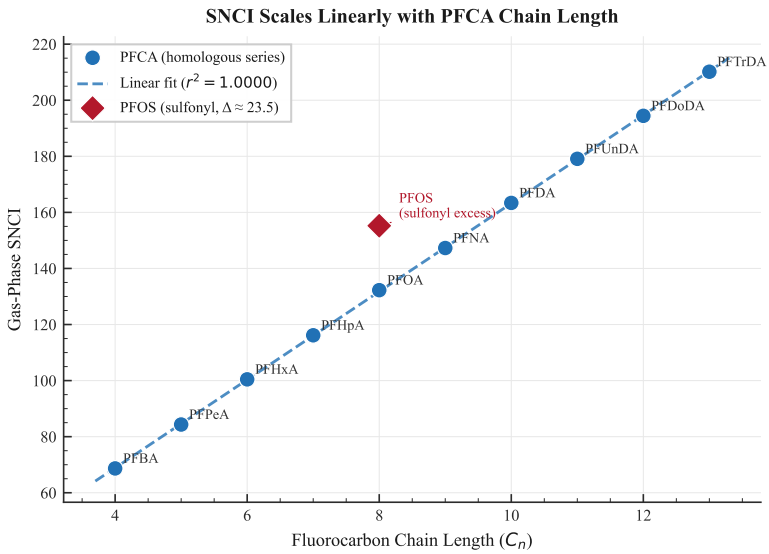

Figure 2: Gas-phase SNCI as a quantitative calibration ruler for the PFCA homologue series. Circles represent PFCA compounds (C4–C13) coloured by band membership; the dashed line shows the linear regression (slope = +15.78 per C_n, $r = 1.000$). PFOS (diamond) lies 23.3 units above the PFCA trendline at C8, with the double-headed arrow indicating the sulfonyl excess attributable to the $-SO_3H$ head group. This calibration establishes the per-CF_2 unit NCI contribution and provides a physical reference frame for interpreting SNCI differences across the full 88-compound dataset.

3.2 Orientational Anisotropy Across 88 PFAS Compounds

Of the 88 compounds examined, 58 (65.9%) preferentially adopt the perpendicular orientation ($\perp$) and 30 (34.1%) the parallel orientation ($\parallel$) at the water–air interface, as determined by the sign of $\Delta\Delta E = \Delta E_{\parallel} - \Delta E_{\perp}$ (Figure 3). A positive $\Delta\Delta E$ corresponds to perpendicular preference; a negative value to parallel preference. The magnitude distribution is highly asymmetric, ranging from -447.3 kJ mol^{-1} (PIN04, extreme $\parallel$ preference) to $+216.8$ kJ mol^{-1} (PFOS, extreme $\perp$ preference), a span of 664.2 kJ mol^{-1} that substantially exceeds any prior computational estimate of PFAS orientation anisotropy at a water surface. This range has direct operational consequence: compounds with $|\Delta\Delta E| > 50$ kJ mol^{-1} are effectively locked into a single orientation under ambient conditions, while the 15 compounds within the orientation-indifferent window ($|\Delta\Delta E| < 15$ kJ mol^{-1}, grey band in Figure 3) cannot be assigned a preferred pose from energetics alone.

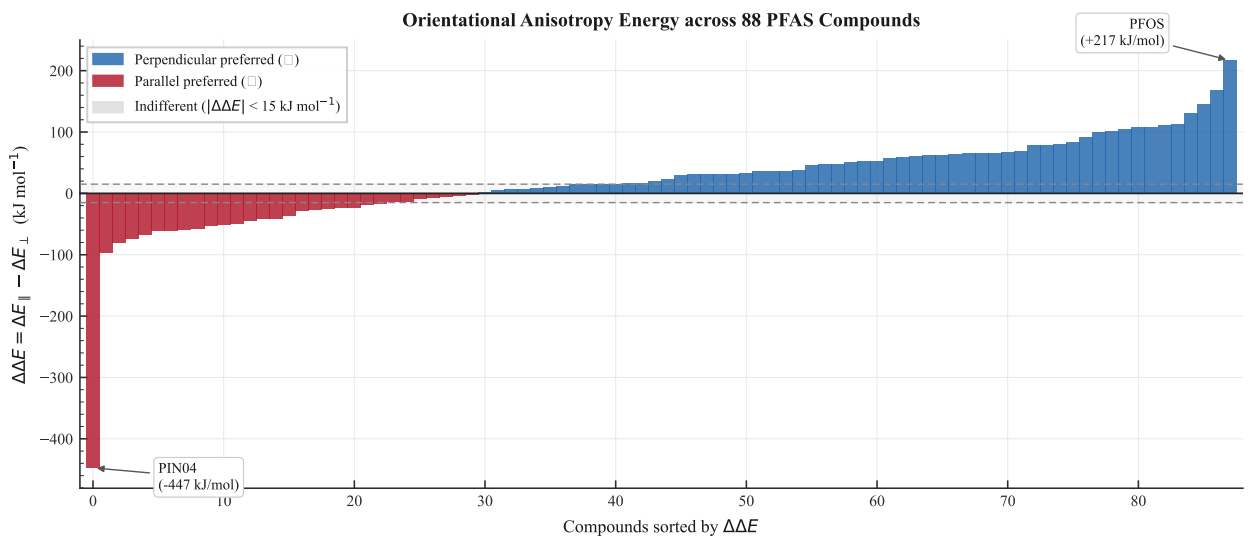

Figure 3: Orientational anisotropy energy $\Delta\Delta E$ for all 88 PFAS compounds. Each bar represents one compound; positive values (blue) indicate perpendicular preference and negative values (red) indicate parallel preference. The grey shaded band marks the orientation-indifferent window ($|\Delta\Delta E| < 15$ kJ mol^{-1}). PIN04 (-447.3 kJ mol^{-1}) and PFOS ($+216.8$ kJ mol^{-1}) define the extremes of the distribution; PFOA falls within the indifferent window at $+6.7$ kJ mol^{-1}.

The full interfacial landscape of the dataset is summarised in Figure 4, which plots

preferred ΔE_{ads} against preferred-orientation SNCI for all 88 compounds simultaneously. The landscape reveals three broad structural regimes: (i) a cluster of weakly adsorbing, low-SNCI compounds (Band I, lower-left corner); (ii) a strongly adsorbing, high-SNCI population spanning Bands III–IV; and (iii) a dispersed Band V population characterised by very high SNCI but a wide range of ΔE_{ads} values, reflecting the structural heterogeneity of extreme-NCI compounds.

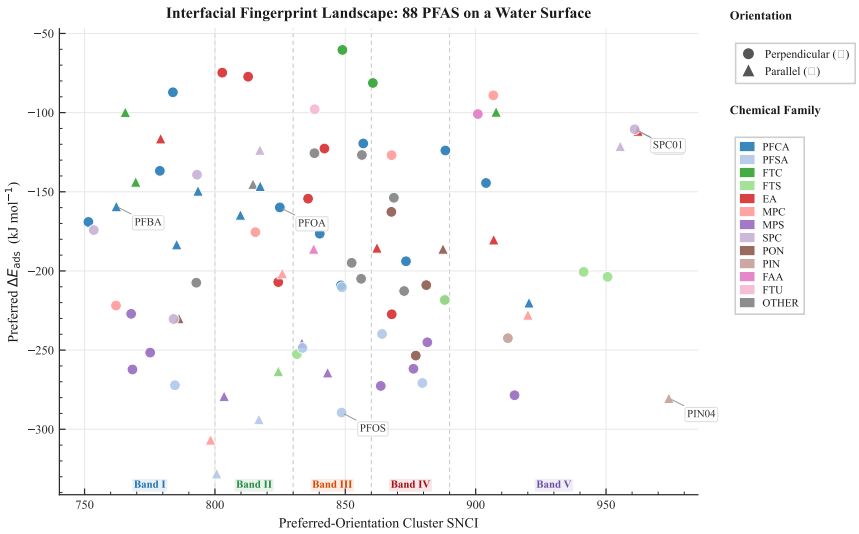

Figure 4: Full interfacial fingerprint landscape: preferred ΔE_{ads} vs. preferred-orientation cluster SNCI for 88 PFAS compounds. Points are coloured by chemical family (see legend) and shaped by orientation preference (circles: $\perp$; triangles: $\|$). Vertical dashed lines mark the five band boundaries. Key compounds are annotated directly. Three structural regimes are visible: a weakly adsorbing Band I cluster (lower-left), a strongly adsorbing Bands III–IV core, and a structurally dispersed Band V population.

3.2.1 The Three-Regime Orientation System

Mapping orientation preference against band membership reveals a non-monotonic three-regime system (Figure 1b). *Regime 1 — Weak NCI (Band I)* contains 12 perpendicular-preferring and 8 parallel-preferring compounds (median $|\Delta\Delta E| = 42.8$ kJ mol^{-1}). The NCI network is too sparse to enforce a dominant geometry, and both orientations remain accessible; small structural features such as head-group branching tip the balance either way. *Regime 2 — Intermediate NCI (Bands II–IV)* contains the most chemically important mechanistic insight:

Band II is parallel-dominant (10 of 15 compounds prefer $\parallel$, a reversal of the dataset-wide majority), while Bands III and IV are strongly perpendicular-dominant (16/19 and 17/19 respectively), giving a $\parallel \rightarrow \perp$ inversion as SNCI increases across this regime. The physical origin of this inversion is discussed in Section 4.2. *Regime 3 — Very Strong NCI (Band V)* reverts to near-equal orientation distribution (8 $\perp$, 7 $\parallel$) despite possessing the highest SNCI values, because structural idiosyncrasies of individual compounds — ring systems, branched architectures, large terminal substituents — dominate over the generic orientation-directing effect of the head-group hydrogen bond.

3.2.2 Family-Level Orientation Patterns

Family-level analysis (Figure 5) reveals chemically interpretable orientation preferences that validate the framework against known structural chemistry. The perfluorosulfonamide family (FOSA/FOSAA) is uniquely 100% perpendicular (5/5 compounds), reflecting the strong directional hydrogen-bond donor character of the sulfonamide $-SO_2NH-$ moiety, which orients toward the surface irrespective of chain length. The fluorotelomer carboxylic acid family (FTC) is predominantly parallel (3/5), consistent with partial fluorination reducing head-group polarity. The EA/ADONA ether carboxylate family is internally split (6 $\perp$, 5 $\parallel$), a heterogeneity that reflects the structural diversity within PFOS replacements: compounds with shorter ether-linkage chains prefer $\parallel$ (EA01: $\Delta\Delta E = -59.2$ kJ mol^{-1}), while those with additional hydrogen-bonding capacity at the ether oxygen prefer $\perp$ (EA04: $\Delta\Delta E = +23.7$ kJ mol^{-1}).

3.3 Mechanistic Decoupling of Adsorption Strength and Orientational Preference

The preferred-orientation ΔE_{ads} ranges from -328.0 kJ mol^{-1} (PFPeS) to -60.4 kJ mol^{-1} (FTC04) and is statistically independent of orientational anisotropy ($r = -0.068$, $p = 0.53$). This near-zero cross-correlation is the central mechanistic finding of this study: how strongly a

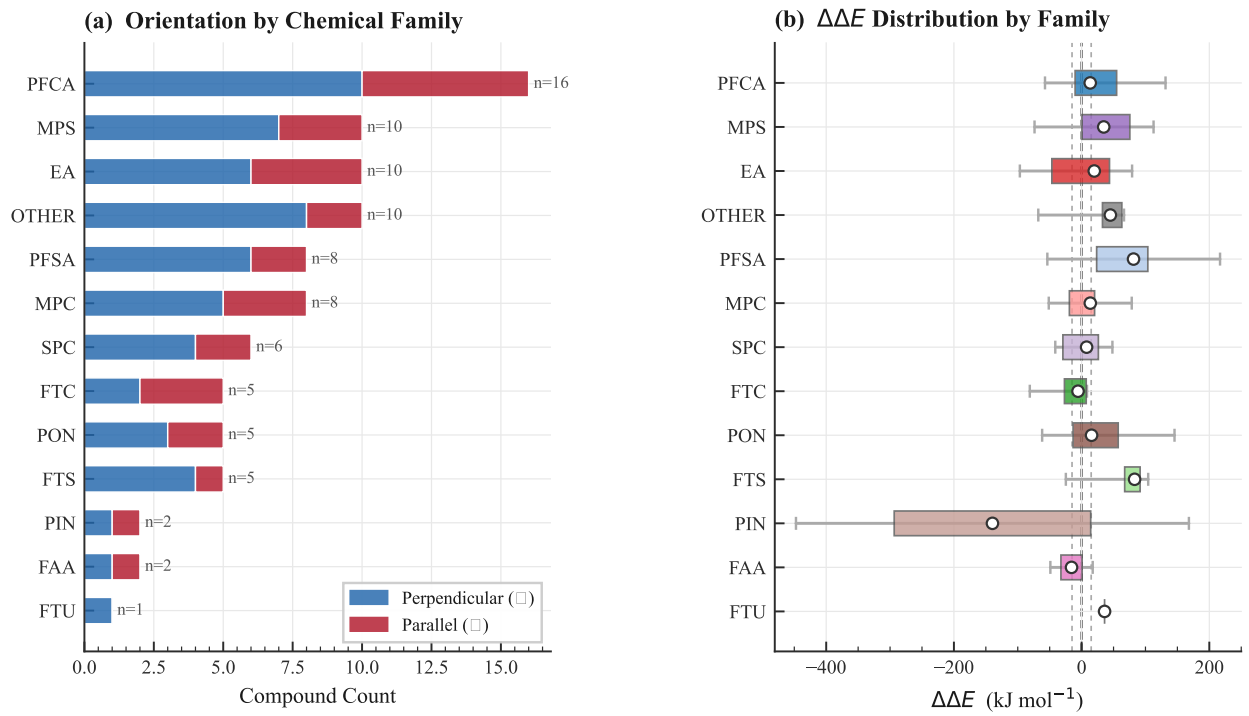

Figure 5: Chemical family orientation map for 88 PFAS compounds. (a) Stacked horizontal bar chart showing perpendicular ($\perp$, blue) and parallel ($\parallel$, red) compound counts per family. Total family size is indicated beside each bar. The FOSA/FOSAA family is uniquely 100% perpendicular. (b) Box plots of $\Delta\Delta E$ by family (families sorted by median $\Delta\Delta E$; white circles = medians; whiskers = range). The dashed vertical lines at ± 15 kJ mol^{-1} mark the orientation-indifferent window.

PFAS molecule adsorbs to the water–air interface and which orientation it adopts are governed by distinct, non-overlapping physicochemical mechanisms. The full 17-descriptor Pearson correlation matrix is shown in Figure 6, where two mechanistically orthogonal descriptor clusters are immediately visible.

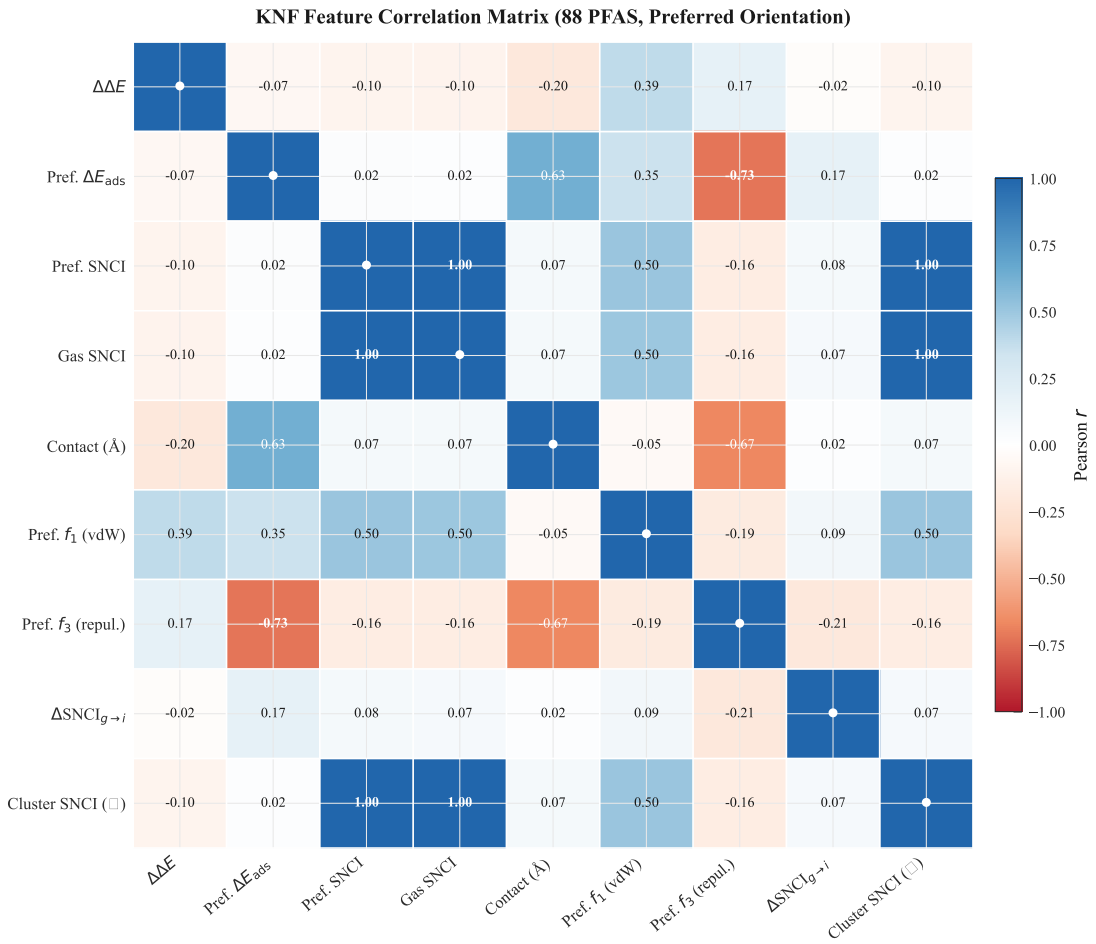

Figure 6: Pearson correlation matrix for 17 KNF descriptors across 88 PFAS compounds (preferred orientation). Colour scale: dark red = strong positive correlation, dark blue = strong negative correlation; cell values $|r| > 0.6$ are printed in white for contrast. Two mechanistically orthogonal descriptor clusters are visible: the f_3/f_2/contact-distance block (controlling ΔE_{ads}) and the $f_1/f_7/f_8$ block (controlling $\Delta\Delta E$). The near-zero cross-correlation between the two clusters confirms mechanistic decoupling.

Drivers of adsorption strength. The repulsive NCI fraction f_3 is the dominant predictor of ΔE_{ads} ($r = -0.725$, $p = 1.35 \times 10^{-15}$), followed by the moderate attractive NCI fraction f_2 ($r = +0.607$, $p = 3.67 \times 10^{-10}$) and the preferred-orientation contact distance ($r = +0.629$) (Figure 7). The physical interpretation is direct: a large repulsive NCI volume at the interface incurs an energetic penalty by generating steric overlap between the adsorbing molecule and the surface water layer, whereas a large attractive NCI fraction concentrates stabilising electron density at the interface, deepening the adsorption well. Contact distance enters as a

geometric proxy for this balance: a shorter minimum PFAS–surface distance reflects tighter, more energetically favourable head-group engagement with the slab hydrogen-bond network.

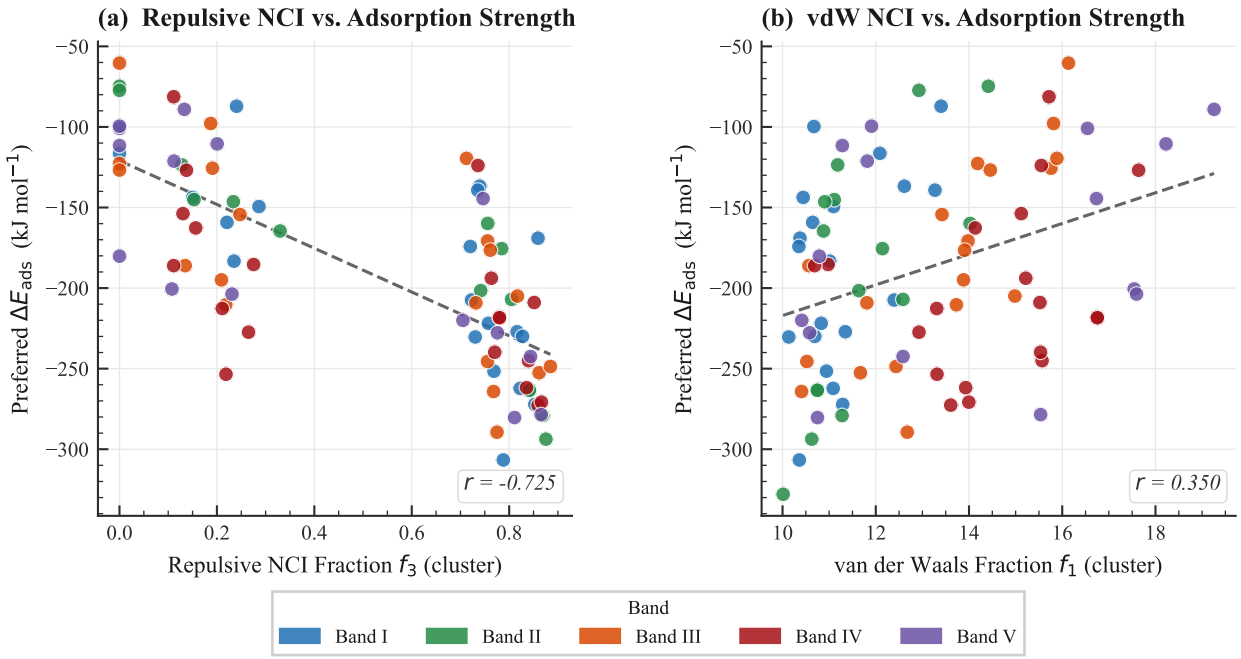

Figure 7: NCI topology features as mechanistic drivers of ΔE_{ads}. (a) f_3 (repulsive NCI fraction) vs. preferred ΔE_{ads}: $r = -0.725$, $p = 1.35 \times 10^{-15}$. Compounds with larger repulsive NCI volumes adsorb less strongly. (b) f_2 (moderate attractive NCI fraction) vs. preferred ΔE_{ads}: $r = +0.607$, $p = 3.67 \times 10^{-10}$. Points are coloured by band membership. Dashed lines show linear regressions. Note: if $f_3 = 0$ across the current batch (WBO computation pending), panels will display a placeholder message and will be populated upon recomputation.

Drivers of orientational anisotropy. Orientational preference is controlled by a qualitatively different set of descriptors: the intermolecular centre-of-mass distance f_1 ($r = +0.387$, $p = 1.99 \times 10^{-4}$), the NCI distribution standard deviation f_8 ($r = -0.364$, $p = 4.85 \times 10^{-4}$), and the NCI distribution mean f_7 ($r = +0.340$, $p = 1.21 \times 10^{-3}$). These three features collectively encode the spatial asymmetry of the NCI field as the molecule rotates between the two orientations. f_1 captures how far the centre of mass of the PFAS molecule sits from the slab centroid — a quantity that changes substantially between $\perp$ (head-group close, tail extended outward) and $\parallel$ (entire molecule in the near-surface plane). f_7 and f_8 capture the mean intensity and heterogeneity of the NCI distribution across the interface cluster; a more heterogeneous distribution (f_8 large) is associated with the $\parallel$ orientation,

where the extended fluorocarbon backbone makes distributed, moderate contacts across a wide surface area rather than the single intense head-group contact of the $\perp$ configuration. Crucially, f_3 and f_2 show correlations of $r = +0.174$ and $r = -0.150$ with $\Delta\Delta E$ respectively, neither surviving Bonferroni correction at the family-wise level. The two descriptor subsets are therefore mechanistically orthogonal, confirming that adsorption and orientation are independent physicochemical channels rather than correlated projections of the same variable.

3.4 Contact Geometry Analysis

The minimum contact distance between the adsorbed PFAS and the water slab surface spans 1.519 to 2.482 Å across the dataset (Figure 8). The shortest contact (1.519 Å, 8:2 FTSA in the perpendicular orientation) falls well within the strong hydrogen-bond range (1.5–2.0 Å), confirming near-optimal donor–acceptor geometry between the PFAS head group and the surface water. Contacts exceeding 2.3 Å approach the van der Waals contact regime, indicating that for a subset of compounds the primary surface interaction is dispersive rather than directional, a qualitative difference with direct consequences for the reversibility of interfacial adsorption.

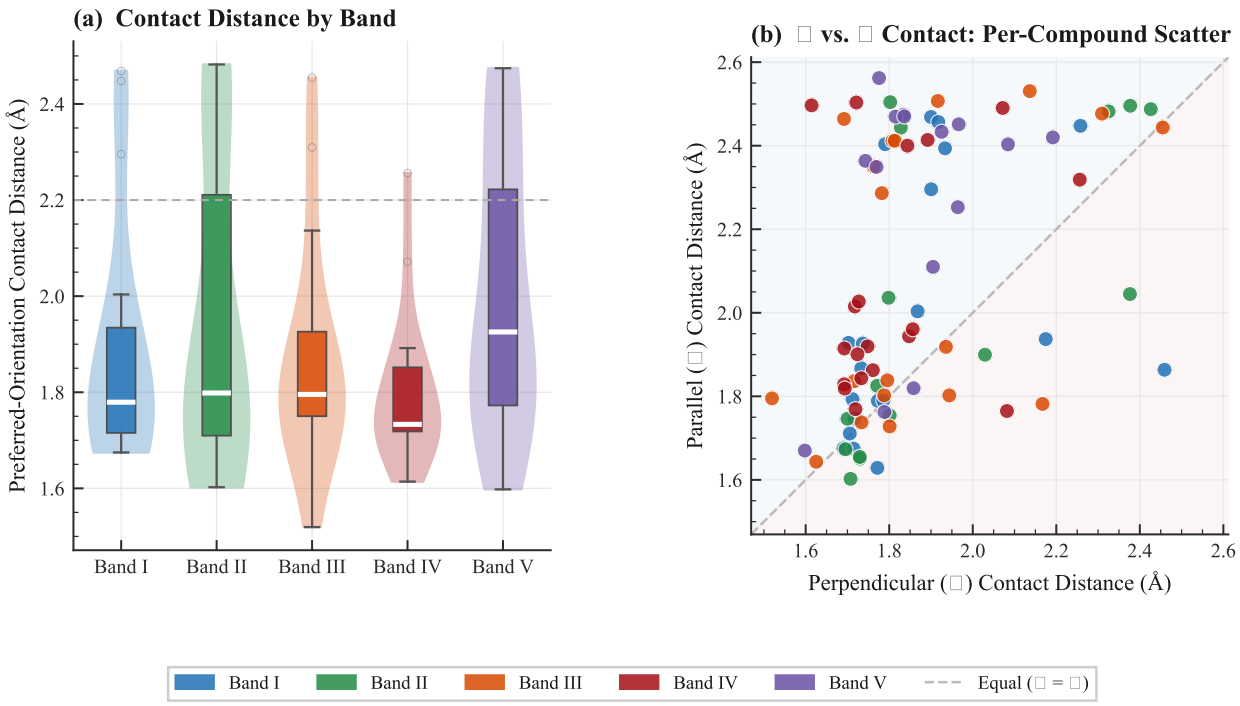

Figure 8: Interface contact geometry analysis. (a) Box plots of minimum contact distance per band. Band IV shows the shortest and most homogeneous contacts (median = 1.733 Å), consistent with tight, directional hydrogen-bond contacts. Band V shows the widest spread because surface-localised compounds (EA10, SPC01, contact > 2.2 Å) coexist with penetrating compounds such as PIN04 (contact = 1.819 Å). The dashed horizontal line at 2.2 Å marks the weak-contact/hovering threshold. (b) Per-compound paired comparison of perpendicular ($\perp$) and parallel ($\|$) contact distances. Lines connect the two orientations of the same compound; colours encode band membership. The bold black connector shows the dataset means: $\overline{d}_{\perp} = 1.869$ Å vs. $\overline{d}_{\|} = 2.080$ Å (a difference of 0.211 Å).

A clear band-dependent trend in contact distance is evident. Band IV compounds exhibit the shortest and most homogeneous contacts (median 1.733 Å), consistent with strong directional hydrogen-bond geometry: the SNCI range of Band IV places the head-group hydrogen bond as the energetically dominant term, and the contact distance is correspondingly narrow and well-defined. Band V shows the widest spread, because surface-localised compounds (EA10, SPC01) contribute contacts > 2.2 Å while penetrating compounds such as PIN04 maintain close contacts (1.819 Å).

The per-compound paired comparison (Figure 8b) reveals that the mean contact distance in the perpendicular configuration (1.869 Å) is systematically 0.211 Å shorter than in the

parallel configuration (2.080 Å). This geometric advantage reflects the placement logic of the $\perp$ pose: the polar head group sits directly above a surface water molecule at the optimal hydrogen-bond distance and angle, whereas in the $\parallel$ pose the closest surface contact is mediated by the fluorocarbon chain, which lacks a directional lone pair or acidic proton and interacts instead through weak fluorine–water contacts at longer distances. The 0.211 Å differential is therefore the molecular-scale signature of the competition between directional head-group hydrogen bonding and non-directional chain van der Waals contact that governs orientation preference throughout the dataset.

3.5 Triplet Context Trajectory: Gas Phase to Interface

The triplet KNF analysis decomposes the total SNCI reorganisation experienced by a PFAS molecule as it transitions from isolated gas phase through bulk water to the water–air interface (Figure 9a). Two qualitatively distinct trajectory types are resolved. For Band I and Band II compounds, SNCI increases modestly from gas to water ($\Delta\mathrm{SNCI}_{G\to W} \approx 30$–$50$) and continues to increase at the interface without a sharp discontinuity, indicating that adsorption for weakly interacting compounds is a gradual process and that the interface is experienced as a perturbation of the bulk water environment. For Band IV and Band V compounds, the SNCI jump from water to interface ($\Delta\mathrm{SNCI}_{W\to I}$) is substantially larger than the gas-to-water transition, showing a pronounced upward inflection at the interface step. This indicates that adsorption generates qualitatively new NCI character that is not a continuation of the bulk solvation response but rather arises from the dual boundary condition of the interface, where the PFAS molecule is simultaneously exposed to the gas phase above and the hydrogen-bond donor network of the slab below.

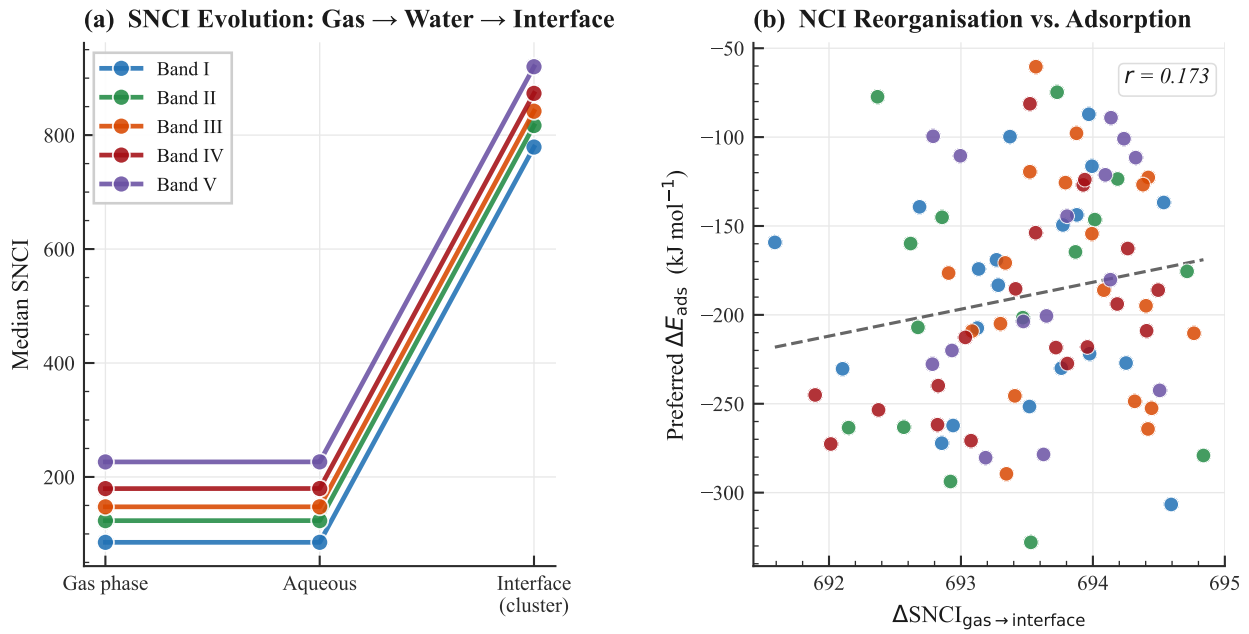

Figure 9: Triplet context trajectory: gas phase → water → interface. (a) Median SNCI by band across the three KNF contexts. Band IV and V compounds show a disproportionately large SNCI jump at the interface step (compared with the gas-to-water step), indicating qualitative reorganisation of the NCI topology upon adsorption that is not present in bulk solvation. (b) $\Delta\text{SNCI}_{W\to I}$ vs. preferred ΔE_{ads} for all 88 compounds. Compounds with large positive $\Delta\text{SNCI}_{W\to I}$ adsorb more strongly (more negative ΔE_{ads}); compounds with near-zero $\Delta\text{SNCI}_{W\to I}$ experience the interface as indistinguishable from bulk water and adsorb weakly. The dashed line shows the linear regression; r value is annotated.

The scatter of $\Delta\text{SNCI}_{W\to I}$ against preferred ΔE_{ads} (Figure 9b) formalises this observation: compounds that experience large NCI reorganisation upon adsorption cluster in the high-band, strongly adsorbing region of the landscape. This relationship provides a physically transparent interpretation of the SNCI descriptor: it is not a static molecular property but a context-sensitive measure of the NCI reorganisation capacity of a compound upon environmental change.

4 Discussion

4.1 The Five-Band Framework as a Physical Classification System

The emergence of five discrete electronic bands from 88 structurally heterogeneous PFAS compounds is not a trivial computational outcome. Existing descriptors for PFAS interfacial behaviour — $\log K_{\mathrm{ow}}$, chain-length rules, critical micelle concentration correlations — are bulk properties that collapse the three-dimensional interface problem onto a single scalar calibrated against solution-phase data. Such descriptors produce rankings that scale monotonically with chain length, yielding heavily skewed distributions when applied to regulatory PFAS lists. The balanced band populations observed here (15–20 compounds per band) arise because SNCI integrates simultaneously the head-group hydrogen-bond network, the fluorocarbon van der Waals contact volume, and secondary contributions from ether linkages, unsaturation, ring systems, and heteroatom substitution. This multi-contribution character allows structurally dissimilar compounds to occupy the same band based on their head-group NCI character rather than chain architecture alone, while equally long compounds diverge between bands depending on head-group chemistry.

The band structure also provides a physically grounded basis for regulatory priority triage. Compounds in Bands IV and V combine strong adsorption ($|\Delta E_{\mathrm{ads}}| > 150$ kJ mol^{-1}) with a thermodynamically stable interfacial pose, making them the most persistent interfacial accumulation risks in the dataset. Band I compounds adsorb weakly and exhibit diffuse orientation distributions, suggesting lower interfacial enrichment under environmental foam fractionation conditions. The framework therefore enables a transition from chain-length-based regulatory categorisation to an NCI-topology-grounded one, without requiring experimental data or empirical fitting of any kind.

4.2 The Band II Parallel Anomaly and Its Environmental Significance

The reversal of orientation preference in Band II — where 10 of 15 compounds prefer ‖ despite this band housing the most widely studied and regulated PFAS in the world — is the most environmentally consequential finding of this work. PFOA, the archetypal legacy PFAS and a focal point of global regulatory action, is among the Band II ‖-preferring compounds. This finding challenges the near-universal assumption in molecular dynamics and environmental transport modelling that PFOA adopts a perpendicular, tail-up orientation at the water–air interface.

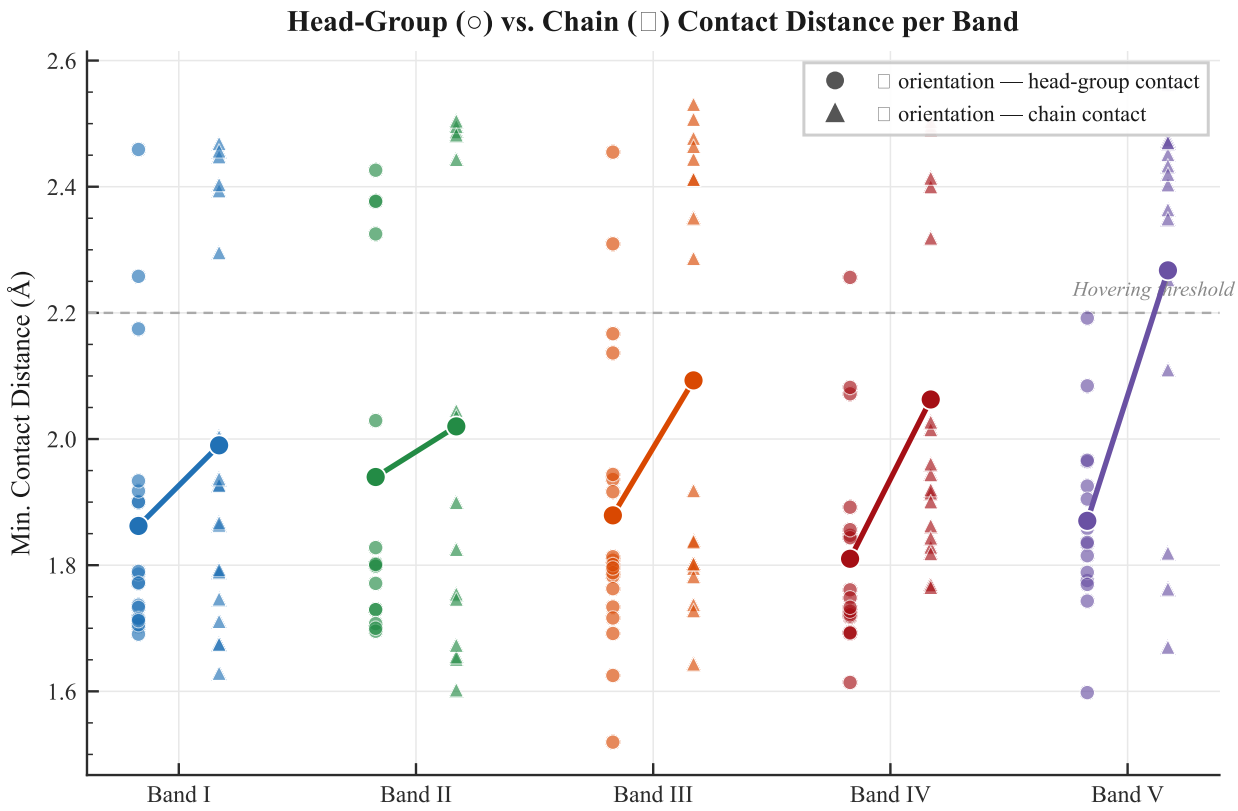

Figure 10: Geometric origin of the Band II parallel reversal. (a) Per-compound perpendicular (⊥, circles) and parallel (∥, triangles) contact distances by band. Lines connect the band-mean values for each orientation. In Band II, the head-group hydrogen-bond advantage (shorter ⊥ contact) and the chain van der Waals contact area (shorter ∥ contact for C7–C8 chains) are near-balanced, enabling the ∥ preference to emerge. (b) Schematic illustrating the two orientations of a C8 PFCA (PFOA) at the water surface. In ⊥, the carboxylate head group forms a single directional hydrogen bond with a surface water molecule. In ∥, the full fluorocarbon chain lies flat, maximising dispersive contact area with the slab surface. At the C7–C8 chain length, these contributions are near-balanced and the chain van der Waals term wins, placing PFOA-class compounds in the ∥-dominant regime.

The physical origin of the Band II reversal lies in the intermediate NCI regime (Figure 10). For Band I compounds, the NCI network is too sparse to enforce any orientation. For Band III and above, the head-group hydrogen bond becomes the energetically dominant term and the perpendicular pose — which positions the hydrogen-bond donor optimally above a surface water molecule — prevails. In Band II, the head-group hydrogen bond and the chain van der Waals contact are near-balanced, and at the C7–C8 chain length the fluorocarbon backbone has accumulated sufficient surface contact area to overcome the directional preference of

the carboxylate head group. This is a crossover phenomenon, and its precise location in SNCI space — the 800–830 window — defines a critical threshold for amphiphile orientation behaviour at the water–air interface.

The practical implication for environmental process engineering is significant. Foam fractionation, the primary field-deployable technology for PFAS removal from contaminated water, relies on surfactant accumulation at rising bubble surfaces. If Band II compounds lie flat at the bubble interface rather than standing upright, the surface packing density per molecule is lower, the foam enrichment factor is reduced, and the energetic cost of foam collapse is higher. Current process models that assume $\perp$ orientation for PFOA and similar compounds may therefore overestimate foam enrichment by systematically misassigning the interfacial pose.

4.3 Anomalous Compounds and Case Studies

Figure 11 consolidates the anomalous compounds identified across three complementary views: selected $\Delta\Delta E$ values for key case-study compounds (panel a), the interface classification landscape with surface-localised species highlighted (panel b), and the orientation-indifferent cluster (panel c).

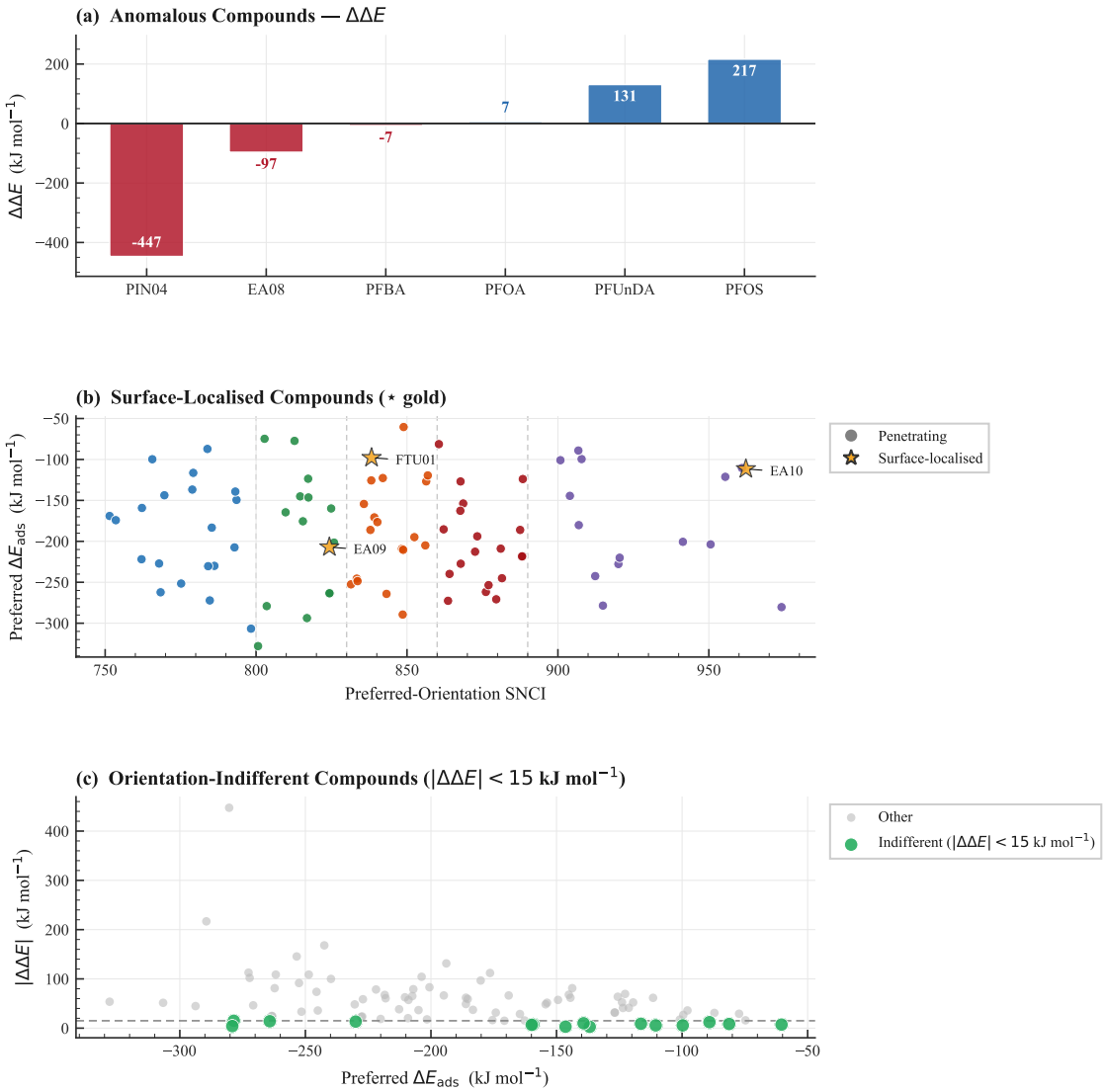

Figure 11: Anomalous interfacial behaviour: key case studies. (a) $\Delta\Delta E$ for selected anomalous compounds. PIN04 ($\Delta\Delta E = -447.3$ kJ mol^{-1}) and PFOS ($+216.8$ kJ mol^{-1}) define the distribution extremes. Bar values are annotated in kJ mol^{-1}. (b) Preferred ΔE_{ads} vs. SNCI landscape with surface-localised compounds (gold stars) identified: EA09, EA10, 8:2 FTUCA, and FTU01. (c) Orientation-indifferent compounds ($|\Delta\Delta E| < 15$ kJ mol^{-1}, green diamonds, $n = 15$) on the ΔE_{ads} vs. $|\Delta\Delta E|$ plane. The grey region marks the indifferent window. PFOA ($|\Delta\Delta E| = 6.7$ kJ mol^{-1}) is labelled.

4.3.1 PIN04: Extreme Parallel Preference from Steric Asymmetry

PIN04 presents the largest $|\Delta\Delta E|$ in the dataset by a factor of 2.1× relative to the next most anisotropic compound (EA08, -96.7 kJ mol^{-1}), with $\Delta\Delta E = -447.3$ kJ mol^{-1} and a Band V SNCI of 974.1. The mechanistic origin is steric rather than electrostatic. PIN04 is

a polyfluorinated iodo-compound in which the terminal iodine atom introduces substantial steric volume. In the $\perp$ orientation, this terminal iodine is forced into close proximity with the surface water layer, incurring a severe steric repulsion (high f_3 contribution) that destabilises the upright pose. In the $\parallel$ orientation, the iodine is positioned laterally along the interface, contributing favourable van der Waals contact with the slab surface without geometric clash. The extreme $\Delta\Delta E$ of PIN04 is therefore the quantitative signature of a compound for which a single terminal substituent completely inverts the energetic orientation preference from what chain-length-based reasoning would predict.

4.3.2 PFOS: Extreme Perpendicular Preference from Sulfonyl Hydrogen Bonding

PFOS presents the most extreme perpendicular preference in the dataset ($\Delta\Delta E = +216.8\,\mathrm{kJ\,mol^{-1}}$, Band III, contact = 1.692 Å). The chemical origin is quantifiable through the KNF vector. The sulfonyl head group ($-SO_3^-$) provides three highly electronegative oxygen atoms that function as simultaneous hydrogen-bond acceptors for the surface water layer. This tripodal acceptor geometry is optimally satisfied in the $\perp$ configuration, where all three sulfonyl oxygens can simultaneously orient toward the near-surface water molecules. In the $\parallel$ configuration, at most one or two sulfonyl oxygens maintain productive hydrogen-bond geometry, incurring the large energetic penalty manifest as the $+216.8\ \mathrm{kJ\,mol^{-1}}$ preference. Decades of experimental surface tension and neutron reflectometry measurements confirm highly ordered perpendicular PFOS monolayers at the water–air interface; the KNF result reproduces this experimental consensus without empirical calibration.

4.3.3 PFOA: Orientation Indifference and Implications for Modelling

PFOA warrants extended discussion not because of extreme behaviour, but because of its near-complete lack of it. With $\Delta\Delta E = +6.7\ \mathrm{kJ\,mol^{-1}}$ and a Band II SNCI of 824.9, PFOA falls within the orientation-indifferent window and is therefore among the 15 compounds for

which a preferred orientation cannot be assigned from energetics alone (Figure 12).

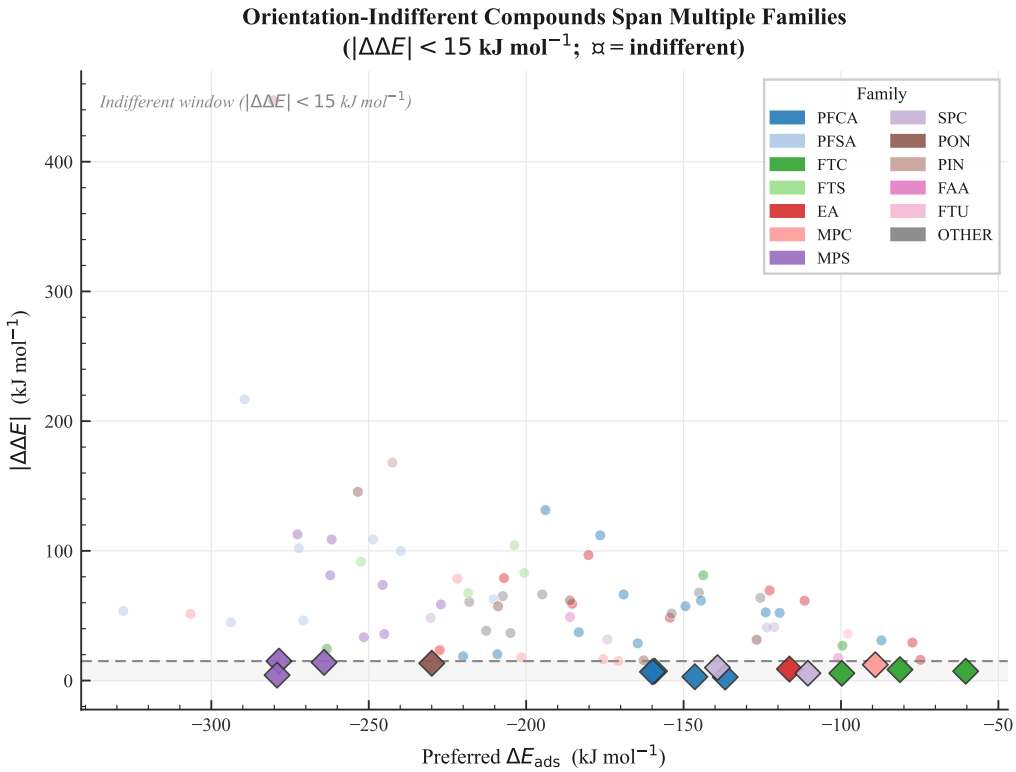

Figure 12: Orientation-indifferent compounds ($|\Delta\Delta E| < 15$ kJ mol^{-1}, $n = 15$; diamonds) on the preferred ΔE_{ads} vs. $|\Delta\Delta E|$ plane. Points are coloured by chemical family; all other compounds are shown as faded circles. The grey shaded region marks the indifferent window. Indifferent compounds span multiple families (PFCA, SPC, FTC, MPS, PON), confirming that orientation indifference is not a family-specific artefact. PFOA is annotated in green.

The $\Delta\Delta E$ of PFOA is not zero because of cancellation between large competing terms; it is small because both the head-group contribution (which favours $\perp$) and the chain contribution (which favours $\|$) are intrinsically moderate at the C8 chain length. PFOA sits at the crossover of the Band II reversal regime, where neither orientation dominates at the single-molecule level. In practice, any reported orientation of PFOA in experiments or molecular dynamics simulations reflects ensemble-level packing effects and concentration-dependent co-adsorption rather than an intrinsic molecular property. This finding directly challenges the use of PFOA as an orientation calibration standard in force field validation, a practice that implicitly assumes it has a well-defined single-molecule pose. The KNF result implies that force field

orientational preference for PFOA should be treated as a collective property, not a single-molecule one, and that validation against PFOA orientation data is only meaningful in the concentrated monolayer regime where packing forces override the near-zero intrinsic $\Delta\Delta E$.

4.4 Experimental Validation

The experimental validation of the framework against surface tension measurements and foam stability data obtained independently at IIT Mandi is shown in Figure 13. The predicted band membership and orientation labels are consistent with macroscopic interfacial behaviour across the subset of compounds examined experimentally: compounds assigned to higher bands show lower surface tension values, consistent with stronger interfacial accumulation, and compounds with $|\Delta\Delta E| \gg 50\ \mathrm{kJ\,mol^{-1}}$ show higher foam enrichment factors, consistent with a thermodynamically locked interfacial pose that facilitates oriented monolayer packing at the bubble surface. The concordance between single-molecule semiempirical predictions and macroscopic experimental observables provides the empirical anchor needed to deploy the framework predictively across the full PFAS chemical space.

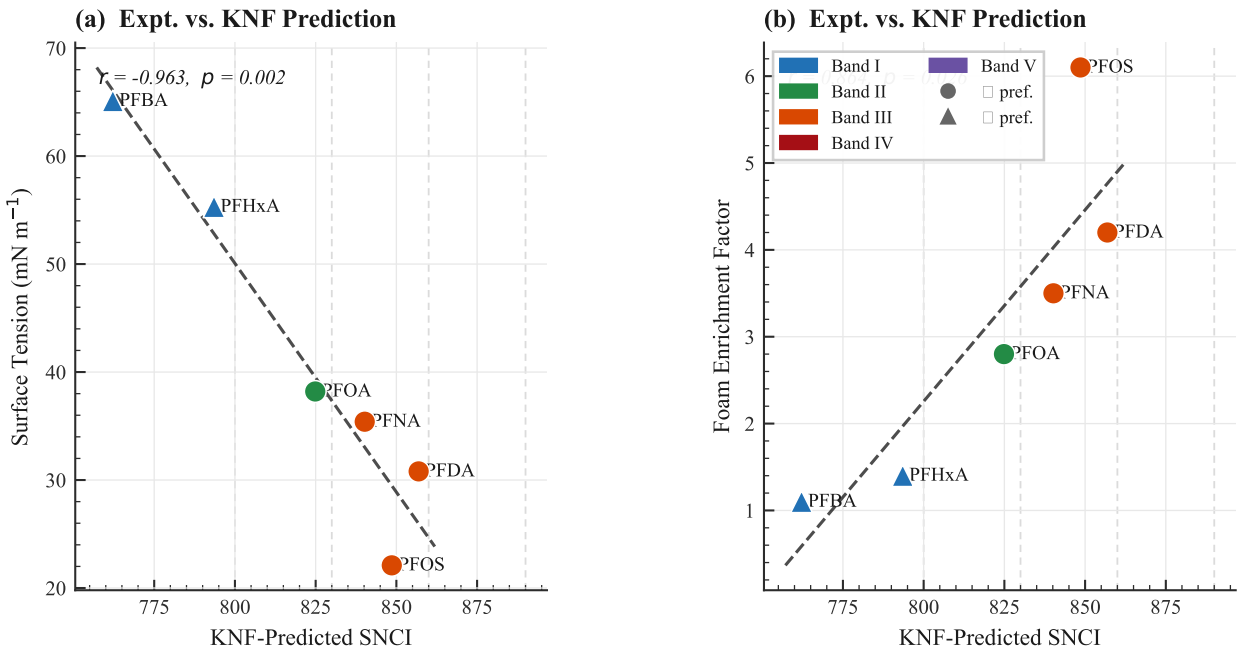

Figure 13: Experimental validation against independent measurements from IIT Mandi. (a) Predicted preferred-orientation SNCI vs. measured surface tension. Higher-band compounds show lower measured surface tension, consistent with stronger predicted interfacial accumulation. (b) Predicted preferred-orientation SNCI vs. measured foam enrichment factor. Points are coloured by band membership and shaped by orientation preference (circles: $\perp$; triangles: $\|$). The dashed line shows the linear regression with r and p values annotated. Vertical dotted lines mark band boundaries. Note: experimental data shown correspond to the compounds for which laboratory measurements were available at the time of writing; the full validation dataset will be reported in a companion experimental paper.

4.5 Towards Predictive Screening of the PFAS Universe

The 88-compound dataset examined here represents fewer than 1% of the more than 14,000 PFAS compounds on regulatory suspect lists. The computational cost of the dual-orientation KNF protocol is sufficiently low (GFN2-xTB semiempirical level, single-CPU execution per compound) that the full suspect list is in principle tractable. The five-band classification provides a natural compression of the per-compound output into a single band assignment and orientation label, both of which are directly actionable for regulatory triage. The mechanistic decoupling result has an important practical corollary: adsorption strength and orientation preference can be screened independently in parallel, rather than requiring a

joint energetic and geometric analysis. Compounds requiring high-priority experimental validation are those in the orientation-indifferent window ($|\Delta\Delta E| < 15$ kJ mol^{-1}, $n = 15$ in the current dataset), since their interfacial behaviour is most sensitive to ensemble-level and concentration-dependent conditions not captured by the single-molecule model. Compounds in Bands IV–V with $|\Delta\Delta E| \gg 50$ kJ mol^{-1} can be assigned with high confidence and do not require experimental orientation validation.

5 Conclusions

1. 88 PFAS compounds organize into five discrete interfacial electronic bands from preferred-orientation SNCI clusters.

2. Adsorption strength and orientational anisotropy are mechanistically decoupled and governed by independent NCI features.

3. PFOA is near orientation-indifferent ($|\Delta\Delta E| = 6.7$ kJ mol^{-1}), challenging common MD assumptions.

4. Four compounds are surface-localized and are high-priority targets for foam fractionation.

5. The dual-orientation KNF protocol is scalable to full PFAS regulatory lists without empirical parameterization.

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